

Homework Set

Emergent Causal Order from Entanglement

This problem set accompanies the paper *Emergent Causal Order from Entanglement: A Pedagogical Introduction*. Every problem is solvable from the paper plus standard undergraduate linear algebra and quantum mechanics.

Conventions. Section, equation, and figure references in this document refer to the paper unless qualified. Problems marked $[\star]$ are slightly harder; problems marked $[\star\star]$ are open-ended or exploratory. Computational problems (Part 8) are optional and require a working build of the `caset.quantum` subsystem.

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1 Partial orders and Kendall correlation

Reference: paper §2.

Problem 1. Let $S = \{a, b, c, d\}$. Define partial order $\preceq_1$ on S to include the strict relations $a \prec_1 b, a \prec_1 c, a \prec_1 d, b \prec_1 d, c \prec_1 d$ (plus reflexivity and transitivity). Define $\preceq_2$ on the same set to include the strict relations $a \prec_2 b, a \prec_2 c, b \prec_2 c, b \prec_2 d, c \prec_2 d$.

- Draw the Hasse diagrams of $\preceq_1$ and $\preceq_2$.
- For each unordered pair $\{x, y\} \subseteq S$ ($x \neq y$), classify the pair as *concordant* (both orders relate it the same way), *discordant* (both orders relate it, opposite ways), *$\preceq_1$ only*, *$\preceq_2$ only*, or *incomparable in both*.
- Compute the Kendall coefficient τ_{12} on the both-comparable subset.

Problem 2. Show formally that majorization is transitive: if $x \preceq y$ and $y \preceq z$ on the simplex Δ_{d-1} , then $x \preceq z$.

Problem 3. $(\star)$ On the set $S = \{1, 2, 3\}$, construct two partial orders such that:

- (a) $\tau_{ab} = -1$;
- (b) τ_{ab} is undefined (because the both-comparable set is empty);
- (c) $\tau_{ab} = 0$.

2 Quantum states and the Schmidt decomposition

Reference: paper §3.

Problem 4. Consider the two-qubit state $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$.

- (a) Write the 2×2 coefficient matrix Ψ of $|\psi\rangle$ in the basis $\{|ij\rangle\}_{i,j \in \{0,1\}}$.
- (b) Compute $\rho_A = \Psi\Psi^\dagger$ and find its eigenvalues.
- (c) Read off the Schmidt spectrum and the entanglement entropy of the cut $\{1\} | \{2\}$.

Problem 5. Let Ψ be the $d_A \times d_B$ coefficient matrix of a pure bipartite state, with SVD $\Psi = USV^\dagger$. Show by direct computation that the matrices $\rho_A = \Psi\Psi^\dagger$ and $\rho_B = \Psi^\dagger\Psi$ have identical nonzero spectra, and that this common spectrum is the squared singular values of Ψ .

Problem 6. (★) For the three-qubit state $|\Psi\rangle = \cos\theta |000\rangle + \sin\theta |111\rangle$, compute the Schmidt spectrum across two different bipartitions:

- (a) the cut $\{1\} | \{2,3\}$;
- (b) the cut $\{1,2\} | \{3\}$.

Are the two spectra equal? Give a one-sentence physical interpretation.

3 Majorization

Reference: paper §4.

Problem 7. For each pair below, decide whether one majorizes the other (and which way) or whether they are incomparable.

- (a) $(0.5, 0.4, 0.1)$ vs $(0.5, 0.3, 0.2)$;
- (b) $(0.7, 0.2, 0.1)$ vs $(0.6, 0.3, 0.1)$;
- (c) $(0.4, 0.3, 0.2, 0.1)$ vs $(0.4, 0.4, 0.1, 0.1)$;
- (d) $(0.5, 0.3, 0.2)$ vs $(0.45, 0.40, 0.15)$.

Problem 8. Using Theorem 4.4 (Nielsen) of the paper, explain why one cannot deterministically convert the Bell pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ into the maximally entangled three-level state $\frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle)$ via LOCC. (Both are pure bipartite states; assume the larger Hilbert space is available on each side.)

Problem 9. (★) Let $H(p) = -\sum_i p_i \log p_i$ denote the Shannon entropy of a probability vector p . Show that majorization implies a reverse ordering on entropy: if $x \preceq y$ then $H(x) \geq H(y)$.

Hint. This is the simplest case of *Schur concavity*. You may use the following fact without proof: for any concave function $f: \mathbb{R} \rightarrow \mathbb{R}$ and any probability vectors $x \preceq y$, the symmetric concave function $\sum_i f(x_i)$ is non-decreasing as the vector becomes more spread out. Specialize to $f(p) = -p \log p$.

4 The lattice Schwinger model

Reference: paper §5.

Problem 10. Consider the hopping operator $h_n = X_n X_{n+1} + Y_n Y_{n+1}$ on two adjacent qubits.

- (a) Write h_n as a 4×4 matrix in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.
- (b) Show that h_n acts as a swap on the $\{|01\rangle, |10\rangle\}$ subspace (with a factor) and annihilates $|00\rangle$ and $|11\rangle$.
- (c) Why is this called a “hopping” term?

Problem 11. Consider an $N = 4$ staggered chain in the configuration where every qubit is in the Z -eigenstate $|0\rangle$ (eigenvalue $+1$). Compute L_n for $n = 1, 2, 3$ (using $L_0 = 0$ and the formula in equation (8) of the paper). Interpret L_n physically as the net charge to the left of link n .

Problem 12. (★) Let $M = \sum_{n=1}^N Z_n$ be the total magnetization (which by the Jordan–Wigner correspondence equals $N - 2 \sum_n c_n^\dagger c_n$, so a conserved M means a conserved total fermion number). Show that M commutes with each of the three pieces of the Hamiltonian: H_{hop} , H_m , and H_E . (For H_{hop} the key intermediate identity is $[Z_n + Z_{n+1}, X_n X_{n+1} + Y_n Y_{n+1}] = 0$; prove it using $[Z_n, X_n] = 2iY_n$, $[Z_n, Y_n] = -2iX_n$.)

5 Lieb–Robinson and causal orders

Reference: paper §6 and §7.

Problem 13. Snapshots are recorded at times $t \in \{0, 0.5, 1.0, 1.5\}$ on a five-site chain (labels $\{1, 2, 3, 4, 5\}$). For each of the following Lieb–Robinson velocities, list every space-time pair $(x_a, t_a) \mapsto (x_b, t_b)$ with $t_a < t_b$ that is $\preceq_{\text{LR}}$ -related (in the site–time sense, before any extension to intervals):

- (a) $v = 0.5$;
- (b) $v = 2$.

You may give the answer as a count plus a brief description of the admissible spatial gaps.

Problem 14. Show that on a regular 1+1D chain with $\preceq_{\text{cs}}$ defined as the time-only order, $\preceq_{\text{LR}} \subseteq \preceq_{\text{cs}}$ for any choice of $v_{\text{LR}} > 0$. Conclude that the Kendall coefficient $\tau(\text{LR}, \text{cs})$ on this geometry is always exactly $+1$.

6 Comparison statistics

Reference: paper §9.

Problem 15. On the set $S = \{1, 2, 3, 4, 5\}$, define partial orders $\preceq_a$ and $\preceq_b$ by:

- $\preceq_a$: $1 \prec_a 2 \prec_a 3, 4 \prec_a 5$;
- $\preceq_b$: $1 \prec_b 3, 2 \prec_b 4 \prec_b 5$.

Take both relations to be reflexive and transitively closed.

- (a) For each of the $\binom{5}{2} = 10$ unordered pairs, classify it as concordant, discordant, $\preceq_a$ only, $\preceq_b$ only, or incomparable in both.
- (b) Compute the four counts $n_{cc}, n_{dd}, n_a^o, n_b^o$.
- (c) Compute τ_{ab} .

7 Interpretation

Reference: paper §10 and §11.

Problem 16. State precisely the difference between the three falsification criteria of Section 7.2 of the paper (strong, weak, and trivial-agreement). For each, give one example of an experimental outcome that would constitute that kind of falsification but not the others.

Problem 17. (★) In Table 1 of the paper, $\tau(\text{maj}, \text{LR})$ decreases monotonically as the Lieb–Robinson velocity v is widened from 0.5 to 16.

- (a) Why is this counterintuitive on first inspection? (What does a naive picture predict the trend should be?)
- (b) What does the observed trend imply about how the new pairs — those admitted into $\preceq_{\text{LR}}$ as the cone widens — relate to $\preceq_{\text{maj}}$?
- (c) Sketch a picture of $\preceq_{\text{maj}}$ that would qualitatively reproduce this trend.

Problem 18. (★★) Suggest two refinements of Hypothesis (H) (Section 7.1 of the paper) that might survive the Phase 5 result on the regular chain. For each refinement:

- (a) State the refined hypothesis precisely.
- (b) Identify the additional measurement or computation needed to test it.
- (c) Predict what outcome would constitute support for the refinement.

There is no unique correct answer; assess your refinements on the criterion that they remain falsifiable and physically motivated.

8 Computational exercises (optional)

Reference: paper §8. Requires `caset.quantum` build.

Problem 19. Run `python examples/quantum/lightcone_vs_majorization.py` from the `caset` repository root. Verify that your output reproduces Table 1 of the paper to within run-to-run noise (Trotter and Krylov-subspace effects can shift τ values by a few times 10^{-4}). Report the wall-clock time on your machine.

Problem 20. (★) Modify the scan to add an intermediate mass regime $m/g = 1.0$ on $N = 10$, $T = 1$. Reproduce the columns of Table 1 for this regime. Compare the resulting $\tau(\text{maj}, \text{LR})$ to the values for $m/g = 0.5$ and $m/g = 5.0$ at the same v . Does the dependence on m/g look monotone, or does the intermediate value lie between? Speculate briefly on a physical reason.